Composition

NMR Metabolomics

Figure 2A in publication

```
print('ME dimensions')
```

```
[1] "ME dimensions"
```

```
MEconc_newlong %>% filter(!is.na(`Sample ID`)) %>% filter(!duplicated(`Sample ID`)) %>% dim
```

```
[1] 110  34
```

```
MEconc_newlong %>%ungroup %>%  filter(!is.na(`Sample ID`)) %>% filter(!duplicated(Participant
```

```
[1] 18 34
```

```
print('PB dimensions')
```

```
[1] "PB dimensions"
```

```
PBconc_new %>% filter(!is.na(Sample)) %>% filter(!duplicated(Sample)) %>% dim
```

```
[1] 43 90
```

```
PBconc_new %>% filter(!is.na(Sample)) %>% filter(!duplicated(Sample)) %>% filter(!duplicated
```

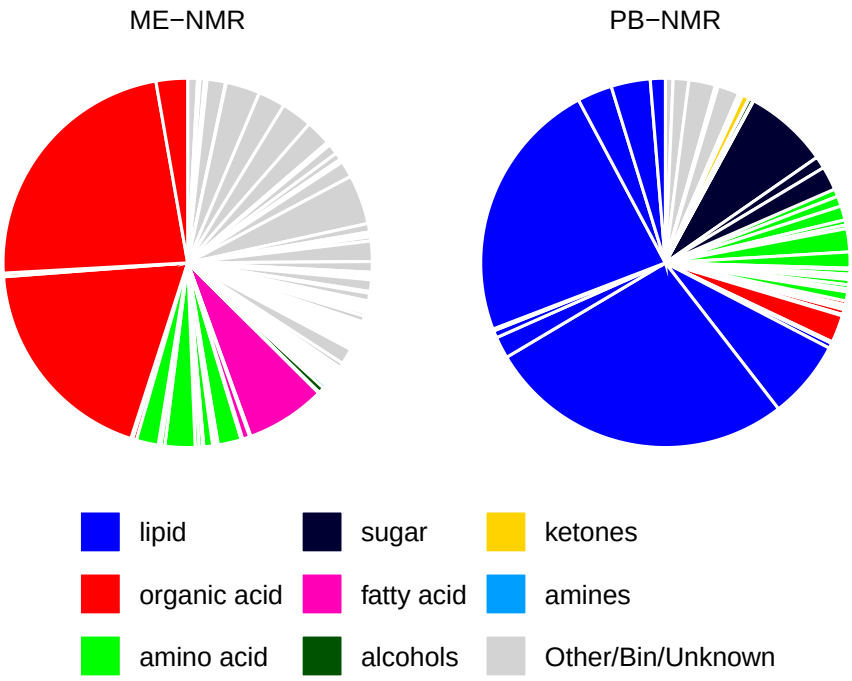
```
[1] 18 90
```

```
df <- bind_rows(
  MEconc_newlong %>%
    filter(! (str_detect(met,'glycerol') | met == 'ERECTIC' | met=='TSP' | str_detect(met,'PC
  PBconc_newlong %>% filter(! (str_detect(met,'glycerol') | str_detect(met,'EDTA') | met ==
) %>%
  filter(!is.na(Participant)) %>%
  group_by(Participant,Cycle,`Cycle day`,`Samples #`) %>%
  # mutate(y = y / sum(y)) %>%
  group_by(type,met, SigMa_class,category) %>%
  dplyr::summarise(mny = sum(y)) %>%
  group_by(type) %>%
  dplyr::mutate(mny = mny / sum(mny)) %>%
  dplyr::mutate(ypos = cumsum(mny) - 0.5*mny,
                xpos = runif(n(),0.5,1.5)) %>%
  ungroup()

df <- df %>% left_join(dflb, by = 'category')
df$lb <- factor(df$lb, levels = dflb$lb[1:9])
```

Composition

```
g_composition <- df %>%
  ggplot(data= ., aes(x = '', y = mny,fill = lb)) +
  geom_bar(stat="identity", width=1, color="white") +
  coord_polar(theta ="y") +
  theme_void() +
  scale_fill_manual(values=dflb$color)   +
  facet_wrap(~type) +
  guides(fill = guide_legend(nrow = 3)) +
  labs(fill = NULL) +
  theme(legend.position = 'bottom' )
g_composition
```



Table

```
df %>%
  dplyr::select(type:mny,lb) %>%
  spread(type,mny) %>%
  reactable()
```

met	SigMa_class	category	lb	ME-NMR	PB-NMR
2-aminobutyric acid	1	organic acid	organic acid		0.00339921690324553
2-hydroxybutyric acid	1	organic acid	organic acid	0.00349821266969007	
2-hydroxyisovaleric acid	1	organic acid	organic acid	0.00109127767200886	
2-oxoisovaleric acid	1	organic acid	organic acid	0.000443115611547181	0.000572493868307065
3-hydroxybutyric acid	1	organic acid	organic acid		0.00116080309033689
acetate	1	organic acid	organic acid	0.188139067793688	0.000686138092429156
acetoacetic acid	1	organic acid	organic acid		0.000493116399027992
acetone	1	ketones	ketones		0.00616754748845636
alanine	1	amino acid	amino acid	0.0203480116943485	0.00681064918721989
alpha-glucose	1	sugar	sugar	0.00143282229197308	

1–10 of 174 rows Previous **1** 2 3 4 5 ... 18 Next

Sampling time, storage time and batch effect correction of ME-NMR data

```
MEconc_newlong <- MEconc_newlong %>%
  mutate(Batch = ifelse(is.na(Batch),3,Batch),
         `Sampling time` = ifelse(is.na(`Sampling time`),'XX',`Sampling time`),
         `Storage time` = ifelse(is.na(`Storage time`),'XX',`Storage time`)) %>%
  group_by(met) %>%
  mutate(logy = residuals(lm(logy~factor(Batch) + factor(`Sampling time`) + factor(`Storage t
  ungroup
```

```
phyITSold <- phyITS
phyXold <- phyX

MEdf <- MEconc_newlong %>%
  dplyr::select(`Sample ID`,`Samples #`, met, logy) %>%
  spread(met,logy) %>% filter(!is.na(`Sample ID`))

MEdfsm <- phyX@sam_data %>%
  data.frame() %>%
  dplyr::select(BRK:Medication.during.sampling) %>%
  left_join(MEdf, by = c('Sample.ID' = 'Sample ID'))

rownames(MEdfsm) <- sample_names(phyXold)

sample_data(phyITS) <- sample_data(phyX) <- sample_data(MEdfsm)

# save.image(file = '~/Dropbox/Bekzod&Isabella/Yongxin/data/QCcorrected_jan2026.Rdata')
```

16s

Primers

Primers for UMI16S:

Primers - Sequence

UMI_27Fa - GTCTCGTGGGCTCGGNNNNNNNNNNNNNNAGAGTTTGATYMTG-GCTYAG UMI_27Fb - GTCTCGTGGGCTCGGNNNNNNNNNNNNNNAGGGTTC-GATTCTGGCTCAG UMI_1540_R - GTCTCGTGGGCTCGGNNNNNNNNNNNNNNNTACG-GYTACCTTGTTACGACT UMI_338Fa - GTCTCGTGGGCTCGGNNNNNNNNNNNNNNNACWC-CTACGGGWGGCAGCAG UMI_338Fb - GTCTCGTGGGCTCGGNNNNNNNNNNNNNNNGACTC-CTACGGGAGGCWGCAG UMI_1391R - GTCTCGTGGGCTCGGNNNNNNNNNNNNNNNGACGGGCG-GTGTGTRCA

Sequencing depth

```
df_ME <- MEconc_newlong %>%
  filter(!duplicated(`Samples #`))

df_descrip <- phyX@sam_data %>% data.frame()
```

```
phyX %>% subset_samples(!is.na(cycle)) %>% sample_sums() %>% summary
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    158   12172   18628   16551   21661   26526
```

```
phyX@sam_data %>% data.frame %>% filter(!is.na(Sample.ID)) %>% filter(!duplicated(Sample.ID))
```

```
[1] 149 162
```

```
phyX@sam_data %>% data.frame %>%  filter(!is.na(Sample.ID)) %>% filter(!duplicated(Participar
```

```
[1]  19 162
```

Table

```
TAXlv <- c()
group_vars_list <- list()
for (i in 1:7){
  txlv <- paste('ta',i,sep = '')
  TAXlv <- c(TAXlv,txlv)
  group_vars_list[[i]] <- TAXlv
}
i <- 0
phyXsummaries <- list()
for (group_vars in group_vars_list) {
  i <- i+1
  txlv <- paste('ta',i,sep = '')
  phyXtaX <- phyX %>%
    subset_samples(!is.na(Sample.ID)) %>%
    transform_sample_counts(function(x) x / sum(x)) %>%
    taxglom(txlv) %>%
    psmelt() %>%
    group_by_at(vars(one_of(group_vars))) %>%
    dplyr::summarise(abu = sum(Abundance),
                     prevalence = sum(Abundance>0) / n()) %>%
    ungroup() %>%
    mutate(abu = abu/ sum(abu),
```

```
            taxlevel = txlv) %>%
    arrange(-abu)
  phyXsummaries[[i]] <- phyXtaX
}

phyXtaX <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  psmelt() %>%
  group_by(ta5,ta6,ta7,OTU) %>%
  dplyr::summarise(abu = sum(Abundance),
                   prevalence = sum(Abundance>0) / n()) %>%
  ungroup() %>%
  mutate(abu = abu/ sum(abu),
         taxlevel = 'ASV') %>%
  arrange(-abu)

phyXsummaries[[8]] <- phyXtaX

fmrt <- list(
  abu = colDef(format = colFormat(percent = TRUE, digits = 1)),
  prevalence = colDef(format = colFormat(percent = TRUE, digits = 1))
)


phyXsummaries[[1]] %>% head %>% reactable(columns = fmrt)
```

ta1	abu	prevalence	taxlevel
K__Bacteria	100.0%	100.0%	ta1
Unassigned	0.0%	28.9%	ta1
K__lambda_standard	0.0%	1.3%	ta1

```
phyXsummaries[[2]] %>% head %>% reactable(columns = fmrt)
```

ta1	ta2	abu	prevalence	taxlevel
K__Bacteria	P__Bacillota	53.9%	100.0%	ta2
K__Bacteria	P__Actinomycetota	26.5%	100.0%	ta2
K__Bacteria	P__Bacteroidota	9.9%	99.3%	ta2
K__Bacteria	P__Pseudomonadota	4.4%	94.6%	ta2
K__Bacteria	P__Fusobacteriota	3.9%	88.6%	ta2
K__Bacteria	P__Mycoplasmatota	0.9%	77.2%	ta2

```
phyXsummaries[[3]] %>% head %>% reactable(columns = fmrt)
```

ta1	ta2	ta3	abu	prevalence	taxlevel
K__Bacteria	P__Bacillota	C__Bacilli	45.0%	100.0%	ta3
K__Bacteria	P__Actinomycetota	C__Actinomycetes	24.1%	100.0%	ta3
K__Bacteria	P__Bacteroidota	C__Bacteroidia	9.9%	99.3%	ta3
K__Bacteria	P__Pseudomonadota	C__Gammaproteobacteria	4.3%	94.0%	ta3
K__Bacteria	P__Fusobacteriota	C__Fusobacteriia	3.9%	88.6%	ta3
K__Bacteria	P__Bacillota	C__Clostridia	3.6%	99.3%	ta3

```
phyXsummaries[[4]] %>% head %>% reactable(columns = fmrt)
```

ta1	ta2	ta3	ta4	abu	prevalence	taxle
K__Bacteria	P__Bacillota	C__Bacilli	O__Lactobacillales	39.9%	100.0%	ta4
K__Bacteria	P__Actinomycetota	C__Actinomycetes	O__Bifidobacteriales	22.6%	100.0%	ta4
K__Bacteria	P__Bacteroidota	C__Bacteroidia	O__Bacteroidales	9.9%	99.3%	ta4
K__Bacteria	P__Bacillota	C__Bacilli	O__Bacillales	5.0%	98.0%	ta4
K__Bacteria	P__Fusobacteriota	C__Fusobacteriia	O__Fusobacteriales	3.9%	88.6%	ta4
K__Bacteria	P__Pseudomonadota	C__Gammaproteobacteria	O__Enterobacterales	3.6%	88.6%	ta4

```
phyXsummaries[[5]] %>% dplyr::select(-ta1,-ta2) %>%  head %>% reactable(columns = fmrt)
```

ta3	ta4	ta5	abu	prevalence	taxlevel
C__Bacilli	O__Lactobacillales	F__Lactobacillaceae	30.3%	100.0%	ta5
C__Actinomycetes	O__Bifidobacteriales	F__Bifidobacteriaceae	22.6%	100.0%	ta5
C__Bacilli	O__Lactobacillales	F__Streptococcaceae	8.3%	96.0%	ta5
C__Bacteroidia	O__Bacteroidales	F__Prevotellaceae	8.3%	98.0%	ta5
C__Fusobacteriia	O__Fusobacteriales	F__Leptotrichiaceae	3.8%	87.9%	ta5
C__Gammaproteobacteria	O__Enterobacterales	F__Enterobacteriaceae	3.6%	87.9%	ta5

```
phyXsummaries[[6]] %>% dplyr::select(-ta1,-ta2,-ta3) %>% head %>% reactable(columns = fmrt)
```

ta4	ta5	ta6	abu	prevalence	taxlevel
O__Lactobacillales	F__Lactobacillaceae	G__Lactobacillus	29.3%	100.0%	ta6
O__Bifidobacteriales	F__Bifidobacteriaceae	G__Gardnerella	22.0%	100.0%	ta6
O__Lactobacillales	F__Streptococcaceae	G__Streptococcus	8.3%	96.0%	ta6
O__Bacteroidales	F__Prevotellaceae	G__Prevotella	5.6%	96.6%	ta6
O__Fusobacteriales	F__Leptotrichiaceae	G__Sneathia	3.8%	87.2%	ta6
O__Enterobacterales	F__Enterobacteriaceae	__	3.6%	86.6%	ta6

```
phyXsummaries[[7]] %>% dplyr::select(-ta1,-ta2,-ta3,-ta4) %>% head %>% reactable(columns = fr
```

ta5	ta6	ta7	abu	prevalence	taxlevel
F__Lactobacillaceae	G__Lactobacillus	__	17.3%	99.3%	ta7
F__Bifidobacteriaceae	G__Gardnerella	__	16.8%	100.0%	ta7
F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	12.0%	99.3%	ta7
F__Streptococcaceae	G__Streptococcus	__	8.3%	96.0%	ta7
F__Bifidobacteriaceae	G__Gardnerella	S__Gardnerella vaginalis ATCC 14018 = JCM 11026	5.0%	84.6%	ta7
F__Enterobacteriaceae	__	__	3.6%	86.6%	ta7

```
phyXsummaries[[8]] %>%  head %>% reactable(columns = fmrt)
```

ta5	ta6	ta7	OTU	abu	prevalence	taxle
F__Lactobacillaceae	G__Lactobacillus	__	sp27	17.3%	99.3%	ASV
F__Bifidobacteriaceae	G__Gardnerella	__	sp5	16.8%	100.0%	ASV
F__Lactobacillaceae	G__Lactobacillus	S__Lactobacillus iners strain DSM 13335	sp1	12.0%	99.3%	ASV
F__Streptococcaceae	G__Streptococcus	__	sp16	8.3%	96.0%	ASV
F__Bifidobacteriaceae	G__Gardnerella	S__Gardnerella vaginalis ATCC 14018 = JCM 11026	sp3	5.0%	84.6%	ASV
F__Enterobacteriaceae	__	__	sp45	3.6%	86.6%	ASV

Composition

Overall distribution

Figure 2A in publication

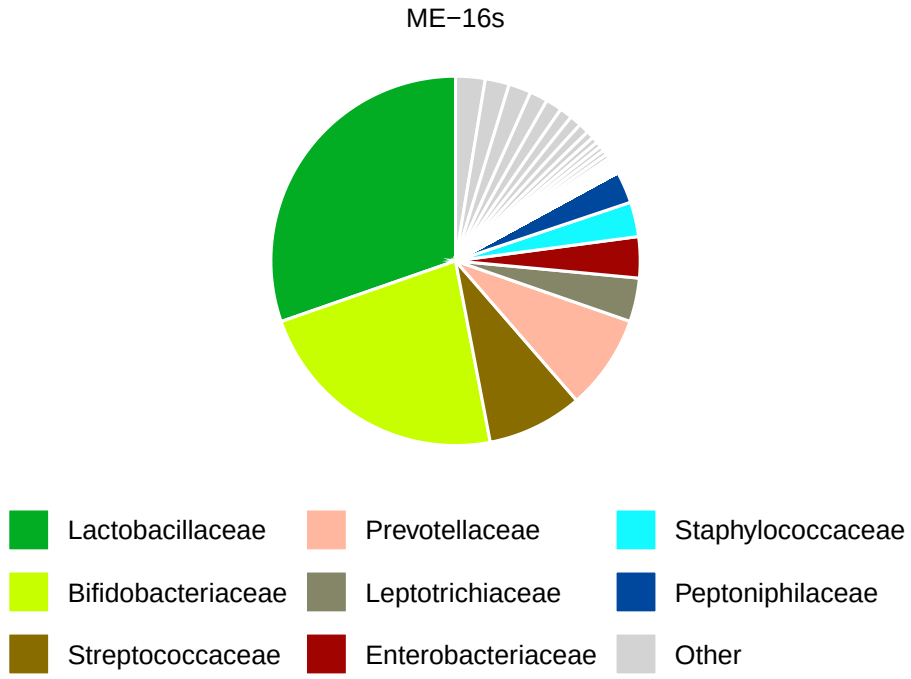
```
b <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  tax_glom('ta5') %>%
  transform_sample_counts(function(x) x/ sum(x)) %>%
  psmelt()

bsum <- b %>%
  group_by(ta2,ta3,ta4,ta5) %>%
  summarise(mn = mean(Abundance)) %>%
  ungroup %>%
```

```
  arrange(-mn) %>%
  mutate(rnk = 1:n(),
         lb = ifelse(rnk<9,substr(ta5,4,30),'Other'),
         lb = factor(lb)) %>%
  mutate(mny = mn / sum(mn)) %>%
  mutate(ypos = cumsum(mn) - 0.5*mn,
         xpos = runif(n(),0.5,1.5))

bsum$lb <- factor(bsum$lb, levels = bsum$lb[1:9])

g_composition16s <- bsum %>%
  mutate(omics = 'ME-16s') %>%
  ggplot(data = ., aes(x = '',y = mn, fill = lb)) +
  geom_bar(stat="identity", width=1, color="white") +
  coord_polar("y", start=0) +
  scale_fill_manual(values=bsum16s$color)   +
  theme_void() +
  guides(fill = guide_legend(nrow = 3)) +
  labs(fill = NULL) +
  # geom_text(data = bsum[1:8,], aes(x = xpos,y = ypos, label = lb), color = "black", size=3)
  theme(legend.position = 'bottom' ) +
  facet_wrap(~omics)
g_composition16s
```



Individual distribution

```
otutb16s <- b %>% left_join(bsum, by = c('ta2','ta3','ta4','ta5'))
participantsort <- otutb16s %>%
  filter(lb=='Lactobacillaceae') %>%
  group_by(Participant) %>%
  dplyr::summarise(ra = mean(Abundance)) %>%
  arrange(-ra) %>%
  mutate(rnk = 1:n())

g2B <- otutb16s %>%
  mutate(Participant = factor(Participant, levels = participantsort$Participant)) %>%
  ggplot(data = ., aes(day,Abundance, fill = lb)) +
  scale_fill_manual(values=bsum16s$color)  +
  geom_bar(stat = 'identity',color = 'grey50')  +
  guides(fill = guide_legend(nrow = 1)) +
  labs(fill = NULL, x = 'Day in Period') +
  facet_grid(cycle~Participant , space = 'free', scales = 'free') +
  theme_bw() +
  theme(legend.position = 'bottom', axis.text.x = element_text(angle = 90))
```

ITS

Primers

Primers for ITS sequencing

ITS1 - GTCTCGTGGGCTCGGTCCGTAGGTGAACCTGCGG ITS4 - GTCTCGTGGGCTCGGTCCTCCGCTTATTGATATGC

Distribution of zero-reads

```
phyyITS_w <- phyITS %>% subset_samples(!is.na(cycle))
table(phyyITS_w@sam_data$Participant,sample_sums(phyyITS_w)>0)
```

```
   FALSE TRUE
A      8    0
B      9    0
E      8    1
G      1    8
J      6    3
K      9    0
M      1    6
N      1    8
O      1    4
P      7    1
Q      7    2
R      9    0
S      2    6
U      9    0
V      3    1
W      7    1
X      3    5
Y      6    3
Z      3    0
```

Of the 19 women, 13 have at least a single sample with ITS sequences.

Sequencing depth

```
v <- phyITS %>% subset_samples(!is.na(cycle)) %>% sample_sums()
```

For all samples

```
summary(v)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      0       0       0     119      54    2116
```

For sample with at least one read

```
table(v>0)
```

```
FALSE  TRUE
  100    49
```

```
summary(v[v>0])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     12      64     178     361     504    2116
```

Table

```
TAXlv <- c()
group_vars_list <- list()
for (i in 1:7){
  txlv <- paste('ta',i,sep = '')
  TAXlv <- c(TAXlv,txlv)
  group_vars_list[[i]] <- TAXlv
}
i <- 0
phyITSsummaries <- list()
phyITSsummaries_df <- data.frame()
for (group_vars in group_vars_list) {
```

```
  i <- i+1
  txlv <- paste('ta',i,sep = '')
  phyITStaX <- phyITS %>%
    subset_samples(!is.na(Sample.ID)) %>%
    transform_sample_counts(function(x) x / sum(x + 1)) %>%
    taxglom(txlv) %>%
    psmelt() %>%
    group_by_at(vars(one_of(group_vars))) %>%
    summarise(abu = sum(Abundance),
              prevalence = sum(Abundance>0) / n()) %>%
    ungroup() %>%
    mutate(abu = abu/ sum(abu),
           taxlevel = txlv) %>%
    arrange(-abu)
  phyITSsummaries[[i]] <- phyITStaX
  phyITSsummaries_df %>% bind_rows(phyITSsummaries_df, phyITStaX)
}

phyITStaX <- phyITS %>%
  subset_samples(!is.na(Sample.ID)) %>%
  transform_sample_counts(function(x) x / sum(x + 1)) %>%
  psmelt() %>%
  group_by(ta4,ta5,ta6,ta7,OTU) %>%
  summarise(abu = sum(Abundance),
            prevalence = sum(Abundance>0) / n()) %>%
  ungroup() %>%
  mutate(abu = abu/ sum(abu),
         taxlevel = txlv) %>%
  arrange(-abu)
phyITSsummaries[[8]] <- phyITStaX

phyITSsummaries[[1]] %>% head %>% reactable(columns = fmrt)
```

ta1	abu	prevalence	taxlevel
K__Eukaryota	100.0%	32.9%	ta1

```
phyITSsummaries[[2]] %>% head %>% reactable(columns = fmrt)
```

ta1	ta2	abu	prevalence	taxlevel
K__Eukaryota	P__Ascomycota	53.6%	32.9%	ta2
K__Eukaryota	P__Basidiomycota	44.5%	26.8%	ta2
K__Eukaryota	__	1.8%	10.7%	ta2

```
phyITSsummaries[[3]] %>% head %>% reactable(columns = fmrt)
```

ta1	ta2	ta3	abu	prevalence	taxlevel
K__Eukaryota	P__Ascomycota	C__Saccharomycetes	42.2%	32.9%	ta3
K__Eukaryota	P__Basidiomycota	C__Agaricomycetes	42.0%	26.2%	ta3
K__Eukaryota	P__Ascomycota	C__Eurotiomycetes	6.4%	22.1%	ta3
K__Eukaryota	P__Ascomycota	C__Pichiomycetes	3.7%	23.5%	ta3
K__Eukaryota	__	__	1.8%	10.7%	ta3
K__Eukaryota	P__Basidiomycota	C__Malasseziomycetes	1.0%	4.7%	ta3

```
phyITSsummaries[[4]] %>% head %>% reactable(columns = fmrt)
```

ta1	ta2	ta3	ta4	abu	prevalence	taxle
K__Eukaryota	P__Ascomycota	C__Saccharomycetes	O__Saccharomycetales	42.2%	32.9%	ta4
K__Eukaryota	P__Basidiomycota	C__Agaricomycetes	O__Agaricales	16.0%	20.8%	ta4
K__Eukaryota	P__Basidiomycota	C__Agaricomycetes	O__Polyporales	15.5%	21.5%	ta4
K__Eukaryota	P__Basidiomycota	C__Agaricomycetes	__	6.2%	13.4%	ta4
K__Eukaryota	P__Ascomycota	C__Eurotiomycetes	O__Eurotiales	5.4%	22.1%	ta4
K__Eukaryota	P__Basidiomycota	C__Agaricomycetes	O__Hymenochaetales	2.9%	10.1%	ta4

```
phyITSsummaries[[5]] %>% dplyr::select(-ta1,-ta2) %>%  head %>% reactable(columns = fmrt)
```

ta3	ta4	ta5	abu	prevalence	taxlevel
C__Saccharomycetes	O__Saccharomycetales	F__Saccharomycetaceae	42.2%	32.9%	ta5
C__Agaricomycetes	__	__	6.2%	13.4%	ta5
C__Agaricomycetes	O__Agaricales	F__Psathyrellaceae	6.0%	14.1%	ta5
C__Eurotiomycetes	O__Eurotiales	F__Aspergillaceae	5.4%	22.1%	ta5
C__Agaricomycetes	O__Agaricales	F__Physalacriaceae	4.5%	7.4%	ta5
C__Agaricomycetes	O__Polyporales	F__Polyporaceae	4.4%	12.1%	ta5

```
phyITSsummaries[[6]] %>% dplyr::select(-ta1,-ta2,-ta3) %>% head %>% reactable(columns = fmrt
```

ta4	ta5	ta6	abu	prevalence	taxlevel
O__Saccharomycetales	F__Saccharomycetaceae	G__Saccharomyces	29.7%	32.9%	ta6
O__Saccharomycetales	F__Saccharomycetaceae	G__Kluyveromyces	12.4%	32.2%	ta6
__	__	__	6.2%	13.4%	ta6
O__Agaricales	F__Psathyrellaceae	G__Coprinellus	5.7%	13.4%	ta6
O__Eurotiales	F__Aspergillaceae	G__Aspergillus	5.2%	21.5%	ta6
O__Polyporales	F__Polyporaceae	G__Trametes	3.5%	10.1%	ta6

```
phyITSsummaries[[7]] %>% dplyr::select(-ta1,-ta2,-ta3,-ta4) %>% head %>% reactable(columns =
```

ta5	ta6	ta7	abu	prevalence	taxlevel
F__Saccharomycetaceae	G__Saccharomyces	S__Saccharomyces_cerevisiae_CBS_1171	16.9%	32.2%	ta7
F__Saccharomycetaceae	G__Saccharomyces	__	12.7%	32.9%	ta7
F__Saccharomycetaceae	G__Kluyveromyces	S__Kluyveromyces_lactis_CBS_683	12.4%	32.2%	ta7
__	__	__	6.2%	13.4%	ta7
F__Psathyrellaceae	G__Coprinellus	__	5.0%	12.8%	ta7
F__Aspergillaceae	G__Aspergillus	S__Aspergillus_glabripes_CCF_5474	3.7%	18.8%	ta7

```
phyITSsummaries[[8]] %>% head %>% reactable(columns = fmrt)
```

ta4	ta5	ta6	ta7	OTU	abu	pre
O__Saccharomycetales	F__Saccharomycetaceae	G__Saccharomyces	S__Saccharomyces_cerevisiae_CBS_1171	sp1	16.9%	
O__Saccharomycetales	F__Saccharomycetaceae	G__Saccharomyces	__	sp9	12.7%	
O__Saccharomycetales	F__Saccharomycetaceae	G__Kluyveromyces	S__Kluyveromyces_lactis_CBS_683	sp7	12.4%	
__	__	__	__	sp5	6.2%	
O__Agaricales	F__Psathyrellaceae	G__Coprinellus	__	sp17	5.0%	
O__Eurotiales	F__Aspergillaceae	G__Aspergillus	S__Aspergillus_glabripes_CCF_5474	sp16	3.7%	

Composition

Overall distribution

Figure 2A in publication

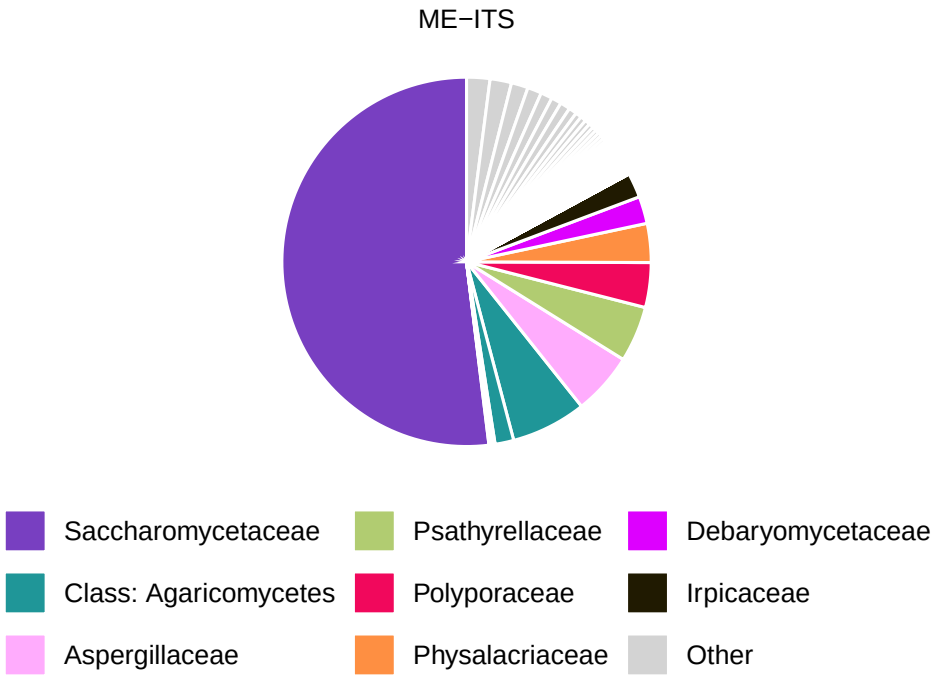
```
b <- phyITS %>%
  subset_samples(!is.na(Sample.ID)) %>%
  tax_glom('ta5') %>%
  transform_sample_counts(function(x) x/ (sum(x) + 1)) %>%
  psmelt()

bsum <- b %>%
  # group_by(ta6) %>%
  group_by(ta3,ta4,ta5) %>%
  summarise(mn = sum(Abundance)) %>%
  ungroup %>%
  arrange(-mn) %>%
  mutate(rnk = 1:n(),
         lb = ifelse(rnk<9,substr(ta5,4,30),'Other'),
         lb = ifelse(is.na(lb),'Other',lb),
```

```
         lb = ifelse(ta5 %in% c('F__','__'),paste('Class', substr(ta3,4,30),sep = ': '),lb))
  ungroup() %>%
  mutate(mn = mn / sum(mn))
bsum$lb <- factor(bsum$lb, levels = bsum$lb[1:9])

g_compositionITS <- bsum %>%
  filter(!is.na(lb)) %>%
  mutate(omics = 'ME-ITS') %>%
  ggplot(data = ., aes(x = '',y = mn, fill = lb)) +
  geom_bar(stat="identity", width=1, color="white") +
  coord_polar("y", start=0) +
  scale_fill_manual(values=bsumITS$color) +
  theme_void() +
  guides(fill = guide_legend(nrow = 3)) +
  labs(fill = NULL) +
  # ggrepel::geom_text_repel(data = bsum, aes(x = xpos,y = ypos, label = lb), color = "black"
  theme(legend.position = 'bottom' ) +
  facet_wrap(~omics)

g_compositionITS
```



Individual distribution

```r
library(ggpattern)

otutbITS <- b %>% left_join(bsumITS, by = c('ta3','ta4','ta5'))
participantsort <- otutbITS %>%
  filter(lb == 'Saccharomycetaceae') %>%
  group_by(Participant) %>%
  dplyr::summarise(ra = mean(Abundance)) %>%
  arrange(-ra) %>%
  mutate(rnk = 1:n())

dfotutbITS <- otutbITS %>%
  group_by(Participant,cycle,day,Sample.ID,lb) %>%
  summarise(Abundance = sum(Abundance)) %>%
  ungroup %>%
  group_by(Sample.ID) %>%
  mutate(Abundance = Abundance / sum(Abundance)) %>%
  mutate(Participant = factor(Participant, levels = participantsort$Participant))

dfmiss <- dfotutbITS %>%
  filter(is.na(Abundance)) %>%
  filter(lb=='Other') %>%
  mutate(Abundance = 1)

g2C <- dfotutbITS %>%
  ggplot(data = ., aes(day,Abundance, fill = lb)) +
  geom_bar(stat = 'identity',color = 'grey50') +
  geom_bar(data = dfmiss, stat = 'identity', fill = 'yellow', alpha = 0.4, color = 'grey50') +
  scale_fill_manual(values=bsumITS$color)  +
  labs(fill = NULL, x = 'Day in Period') +
  theme_bw() +
  facet_grid(cycle~Participant) +
  guides(fill = guide_legend(nrow = 1)) +
  theme(legend.position = 'bottom', axis.text.x = element_text(angle = 90))
g2C
```

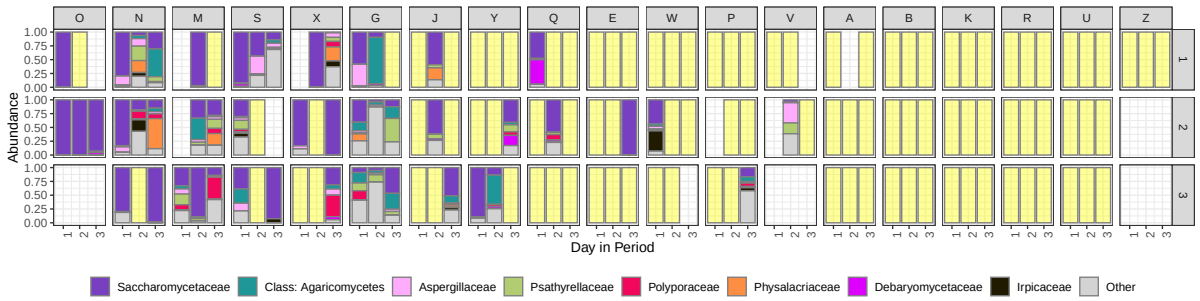
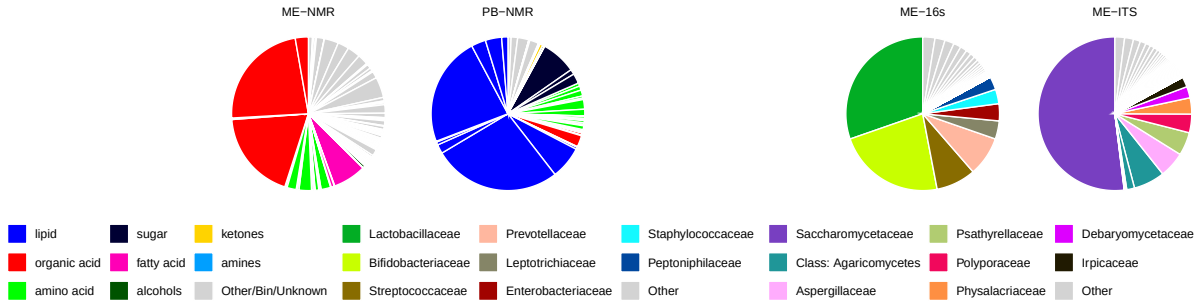


Figure 2 in publication

```
layout2 <- "
AABC
"

all_comp <- g_composition + g_composition16s + g_compositionITS +
  plot_layout(design = layout2,guides = "collect") & theme(legend.position = 'bottom',
                                                           legend.direction = 'horizontal')

all_comp
```




```
ggsave('FIG2A_composition.pdf',plot = all_comp, width = 14)
ggsave('FIG2B_16s_composition.pdf',plot = g2B, width = 14, height = 4)
ggsave('FIG2C_ITS_composition.pdf',plot = g2C, width = 14, height = 4)
```

Trajectories over time

ME

```r
MEconc_newlong %>%
  mutate(cycle_day = (as.numeric(Cycle)-1)*5 + `Cycle day`) %>%
  ggplot(data = ., aes(cycle_day,logy, group = Participant, color = category)) +
  geom_point() + geom_line() +
  theme(legend.position = 'bottom') +
  facet_wrap(~met, ncol = 5, scales = 'free')
```


2−hydroxybutyric acid
2−hydroxyisovaleric acid
2−oxoisovaleric acid
acetate
alanine
alpha−glucose
aspartate
bin−1
bin−1,2−propanediol
bin−2
bin−3
bin−3−hydroxybutyrate
bin−4
bin−aspartate−asparagine
bin−ethanol
bin−fucose−arabinose
bin−isoleucine−lysine
bin−leucine−isoleucine
bin−leucine−lysine−arginine
bin−proline
bin−proline−glutamate
bin−propionate 1_2
bin−propionate−unk
bin−propionate−valine
bin−sucrose
bin−tryptophan
bin−tryptophan−tyramine
bin−tyramine
bin−tyramine−unk48
bin−unk15_1−acetone
bin−unk18−glutamate
bin−unk2
bin−unk21−aspartate
bin−unk29−asparagine
bin−unk47−tryptophan
bin−unk48
bin−unk8−isoleucine
bin−valine−acetoacetate
butyrate
choline
creatine
creatinine
dimethyl sulfone
dimethylamine
ERECTIC
ethanol
formate
fumarate
gamma−aminobutyric acid
glutamate
glycerol
glycine
hypoxanthine
isobutyrate
isoleucine
lactate
leucine
lysine
methanol
methylamine
phenylalanine
propionate
succinate
threonine
trimethylamine
tryptophan
TSP
unk1
unk10
unk11
unk12
unk13_1
unk13_2
unk14_1
unk14_2
unk14_3
unk15_2
unk15_3
unk16
unk17
unk18_2
unk19
unk20
unk21
unk22
unk23
unk24
unk25
unk26
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unk49
unk5
unk50
unk51
unk52
unk53
unk54
unk55
unk56
unk57
unk58
unk59
unk6
unk60
unk7
unk9
uracil
uridine
valine
logy
cycle_day
category
alcohols
amines
amino acid
amino acid derivatives
bin
fatty acid
glycerolipids
internal signal
organic acid
purines and purine derivatives
pyrimidine nucleosides
pyrimidines and pyrimidine derivative
quaternary ammonium salts
sugar
sulfones
unknown

16s

```
otutb16s %>%
  ggplot(data = ., aes(cycle_day,Abundance, group = Participant, color = lb)) +
  geom_point() +
  geom_line() +
  scale_y_log10() +
  theme(legend.position = 'bottom') +
  facet_wrap(~OTU + ta5, ncol = 5)
```


Abundance
cycle_day
lb
Lactobacillaceae
Streptococcaceae
Leptotrichiaceae
Staphylococcaceae
Other
Bifidobacteriaceae
Prevotellaceae
Enterobacteriaceae
Peptoniphilaceae

ITS

```
otutbITS %>%
  ggplot(data = ., aes(cycle_day,Abundance, group = Participant, color = lb)) +
  geom_point() +
  geom_line() +
  scale_y_log10() +
  theme(legend.position = 'bottom') +
  facet_wrap(~OTU + ta5, ncol = 5)
```


Abundance
cycle_day
lb
Saccharomycetaceae
Class: Agaricomycetes
Aspergillaceae
Psathyrellaceae
Polyporaceae
Physalacriaceae
Debaryomycetaceae
Irpicaceae
Other

Effect of day-in-cycle

Vulcano plot

Figure 3B in publication

```
tidyall <- function(m0) {
  r <- bind_rows(
    m0 %>% anova() %>% tidy,
    m0 %>% ranova() %>% tidy,
    m0 %>% tidy)
  return(r)
}

tidyall <- function(fit, data) {
  fit@call$data <- as.data.frame(data)

  bind_rows(
    broom::tidy(anova(fit)) %>%
      mutate(test = "fixed"),

    broom::tidy(lmerTest::ranova(fit)) %>%
      mutate(test = "random"),

    broom.mixed::tidy(fit) %>%
      mutate(test = "coefficient")
  )
}

pc <- 1e-5

tb16s <- phyX %>%
  subset_samples(!is.na(Sample.ID)) %>%
  filter_taxa(function(x) sum(x > 0) > 4, TRUE) %>%
  transform_sample_counts(function(x) x / sum(x)) %>%
  psmelt() %>%
  as.data.frame() %>%
  mutate(lograbu = log(Abundance + pc)) %>%
  group_by(ta3, ta4, ta5, ta6, ta7, OTU) %>%
  group_modify(~ {
    dat <- as.data.frame(.x)
```

```
    fit <- lmerTest::lmer(
      lograbu ~ factor(day) +
        (1 | Participant) +
        (1 | cycle_part),
      data = dat
    )
    fit@call$data <- dat

    # broom::tidy(lmerTest::ranova(fit))
    tidyall(fit, dat)
  }) %>%
  ungroup()



tbITS <- phyITS %>%
  subset_samples(!is.na(Sample.ID)) %>%
  filter_taxa(function(x) sum(x>0)>4, TRUE) %>%
  psmelt() %>% data.frame() %>%
  mutate(lograbu = log(Abundance + pc)) %>%
  group_by(ta3,ta4,ta5,ta6,ta7, OTU) %>%
  group_modify(~ {
    dat <- as.data.frame(.x)

    fit <- lmerTest::lmer(
      lograbu~ factor(day) +
        (1 | Participant) +
        (1 | cycle_part),
      data = dat
    )
    fit@call$data <- dat

    # broom::tidy(lmerTest::ranova(fit))
    tidyall(fit, dat)
  }) %>%
  ungroup()




mME <- MEconc_newlong %>%
  mutate(cycle_part = paste(Participant, Cycle)) %>%
```

```
  group_by(met,category) %>%
  group_modify(~ {
    dat <- as.data.frame(.x)

    fit <- lmerTest::lmer(
      logy~ factor(`Cycle day`) + (1|`Participant`) + (1|cycle_part),
      data = dat
    )
    fit@call$data <- dat
    tidyall(fit, dat)
  }) %>%
  ungroup()



mME_pool <- MEconc_newlong %>%
  filter(is.na(`Sample ID`)) %>%
  filter(Cycle=='O') %>%
  group_by(met,category) %>%
  summarise(sd(logy))
```

```
dfplot <- bind_rows(mME %>% mutate(omics = 'ME-NMR') %>%
                      filter(!str_detect(met,'PC')) %>%
                      filter(!str_detect(met,'glycerol')) %>%
                      filter(!str_detect(met,'TSP')) %>%
                      filter(!str_detect(met,'ERECTI')) %>%
                      left_join(dflb, by = 'category'),
                    tbITS %>% mutate(omics = 'ME-ITS', met = OTU) %>% left_join(bsumITS, by =
                    tb16s %>% mutate(omics = 'ME-16s',met = OTU) %>% left_join(bsum16s, by =
  filter(str_detect(term,'day'))  %>%
  filter(!is.na(estimate)) %>%
  filter(parse_number(term)==2) %>%
  mutate(omics = factor(omics, levels = c('ME-NMR','ME-16s','ME-ITS')))

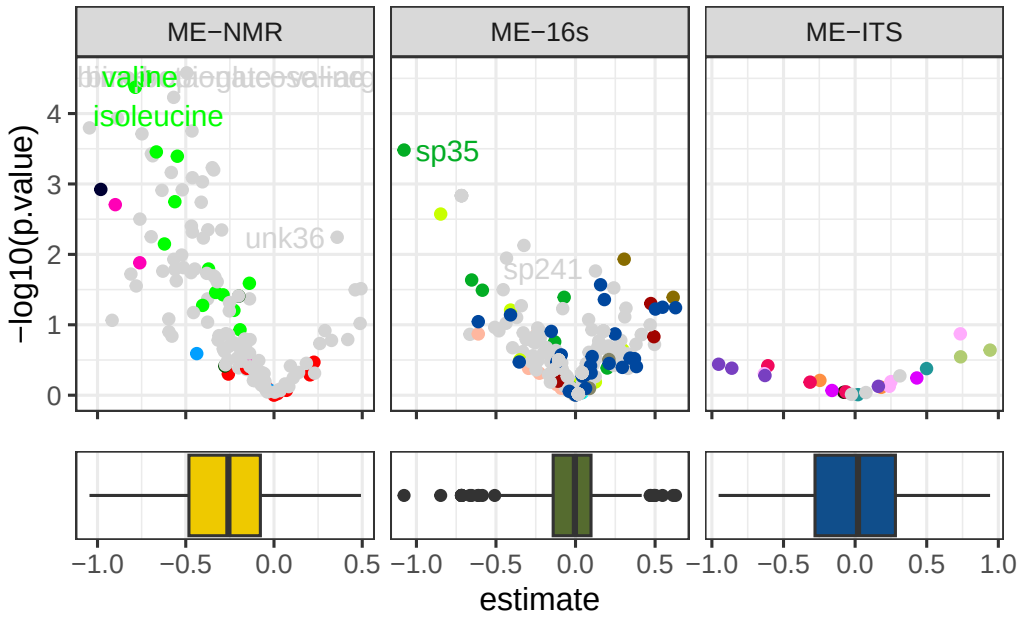
g_box <- dfplot %>%
  ggplot(data =., aes(estimate, fill = omics)) +
  geom_boxplot()+
  facet_wrap(~omics, scales = 'free_x') +
  scale_fill_manual(values = colorcodes, breaks = colorcodesnames) +
  theme_bw() +
  theme(legend.position = 'none',
        strip.text = element_blank(),
```

```
        strip.background = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank(),
        panel.grid.major.y = element_blank(),
        panel.grid.minor.y = element_blank())


g_vulc <- dfplot %>%
  ggplot(data =., aes(estimate,-log10(p.value), label = met, color = lb)) +
  geom_point() +
  facet_wrap(~omics, scales = 'free_x') +
  scale_color_manual(values = bind_rows(dflb, bsum16s,bsumITS)$color,
                     breaks = bind_rows(dflb, bsum16s,bsumITS)$lb)   +
  ggrepel::geom_text_repel(data = dfplot[dfplot$p.value<0.01,]) +
  theme_bw() +
  theme(legend.position = 'none', axis.title.x = element_blank(), axis.text.x = element_blank

library(patchwork)
layout <- "
A
A
A
A
B
"

g_day2 <- g_vulc + g_box+
  plot_layout(design = layout)
g_day2
```




```
ggsave('FIG3B_vulcano.pdf',plot = g_day2, width = 8)
```

Tables

NMR

```
fmrt <- list(
  # estimate = colDef(format = colFormat(scientific = TRUE, digits = 2)),
  estimate = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  ),

  std.error = colDef(
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  ),

  p.value = colDef(
```

```
    cell = function(value) {
      if (is.na(value)) return("")
      formatC(value, format = "e", digits = 2)
    }
  )
)


dfplot %>%
  ungroup %>%
  filter(omics=='ME-NMR') %>%
  filter(!duplicated(met)) %>%
  arrange(p.value) %>%
  dplyr::select(met,category,estimate,std.error,p.value) %>%
  reactable(columns = fmrt) %>%
  add_subtitle("NMR metabolomics - Effect of day 1 vs 2. Effect-sizes and inference used in t
```

NMR metabolomics - Effect of day 1 vs 2. Effect-sizes and inference used in the vulcano-plots, sorted by inference

met	category	estimate	std.error	p.value
bin-beta-glucose-arginine-phenylalanine-histidine-betaine-taurine	bin	-4.94e-01	1.12e-01	2.64e-05
valine	amino acid	-7.12e-01	1.62e-01	3.02e-05
bin-propionate-valine	bin	-8.25e-01	1.88e-01	3.12e-05
isoleucine	amino acid	-7.86e-01	1.80e-01	4.24e-05
bin-valine-acetoacetate	bin	-5.68e-01	1.35e-01	5.89e-05
bin-tyramine-unk48	bin	-8.87e-01	2.20e-01	1.17e-04
bin-unk48	bin	-1.05e+00	2.65e-01	1.60e-04
bin-unk21-aspartate	bin	-4.65e-01	1.19e-01	1.77e-04
unk44	unknown	-7.48e-01	1.90e-01	1.94e-04
phenylalanine	amino acid	-6.67e-01	1.79e-01	3.51e-04

1–10 of 135 rows Previous **1** 2 3 4 5 ... 14 Next

16s microbiome

```
dfplot %>%
  ungroup %>%
  filter(omics=='ME-16s') %>%
  left_join(phyXsummaries[[8]], by = c('met' = 'OTU')) %>%
  filter(!duplicated(met)) %>%
  arrange(p.value) %>%
  dplyr::select(ta4:OTU,estimate,std.error,p.value) %>%
  reactable(columns = fmrt) %>%
  add_subtitle("16s sequencing - Effect of day 1 vs 2. Effect-sizes and inference used in the
```

16s sequencing - Effect of day 1 vs 2. Effect-sizes and inference used in the vulcano-plots, sorted by inference

ta4	ta5.x	ta6.x	ta7.x	OTU	estimate	st
O__Lactobacillales	F__Lactobacillaceae	G__Limosilactobacillus	__	sp35	-1.08e+00	2
__	__	__	__	sp97	-7.17e-01	2
O__Bifidobacteriales	F__Bifidobacteriaceae	G__Bifidobacterium	__	sp46	-8.47e-01	2
O__Eubacteriales	F__Oscillospiraceae	G__Mageeibacillus	S__Mageeibacillus indolicus strain 0009-5	sp241	-3.24e-01	1
O__Actinomycetales	F__Actinomycetaceae	G__Pauljensenia	S__Pauljensenia hongkongensis strain HKU8	sp242	-4.34e-01	1
O__Lactobacillales	F__Streptococcaceae	G__Streptococcus	S__Streptococcus urinalis strain 2285-97	sp194	3.06e-01	1
O__Actinomycetales	F__Actinomycetaceae	G__Arcanobacterium	S__Arcanobacterium urinimassiliense strain Marseille-P3248	sp128	1.25e-01	5
O__Lactobacillales	F__Lactobacillaceae	G__Ligilactobacillus	__	sp39	-6.53e-01	2
O__Tissierellales	F__Peptoniphilaceae	G__Finegoldia	S__Finegoldia magna strain JCM 1766	sp135	1.57e-01	7
O__Eubacteriales	F__	G__Gemmiger	S__Gemmiger qucibialis	sp83	2.40e-01	1

ITS Fungome

```
dfplot %>%
  ungroup %>%
  filter(omics=='ME-ITS') %>%
  left_join(phyITSsummaries[[8]], by = c('met' = 'OTU')) %>%
  filter(!duplicated(met)) %>%
  arrange(p.value) %>%
  dplyr::select(ta4.x:OTU,estimate,std.error,p.value) %>%
  reactable(columns = fmrt) %>%
  add_subtitle("ITS sequencing - Effect of day 1 vs 2. Effect-sizes and inference used in the
```

ITS sequencing - Effect of day 1 vs 2. Effect-sizes and inference used in the vulcano-plots, sorted by inference

ta4.x	ta5.x	ta6.x	ta7.x	OTU	estimate	st
O__Eurotiales	F__Aspergillaceae	G__Penicillium	__	sp10	7.35e-01	4
O__Agaricales	F__Psathyrellaceae	G__Coprinellus	__	sp17	9.42e-01	7
O__Agaricales	F__Psathyrellaceae	G__Coprinellus	S__Coprinellus_aureodisseminatus_HMJAU_67119	sp29	7.37e-01	6
O__Saccharomycetales	F__Saccharomycetaceae	G__Kluyveromyces	S__Kluyveromyces_lactis_CBS_683	sp7	-9.53e-01	1
O__Polyporales	F__Polyporaceae	G__Trametes	S__Trametes_versicolor_CFMR_FP-135156-Sp	sp14	-6.09e-01	6
O__Saccharomycetales	F__Saccharomycetaceae	G__Saccharomyces	S__Saccharomyces_cerevisiae_CBS_1171	sp1	-8.61e-01	1
O__Agaricales	__	__	__	sp15	4.99e-01	6
O__Eurotiales	F__Aspergillaceae	G__Aspergillus	S__Aspergillus_glabripes_CCF_5474	sp16	-6.34e-01	9
O__Saccharomycetales	F__Saccharomycetaceae	G__Saccharomyces	__	sp9	-6.29e-01	9
O__Malasseziales	F__Malasseziaceae	G__Malassezia	S__Malassezia_restricta_CBS_7877	sp67	3.13e-01	4

1–10 of 23 rows Previous **1** 2 3 Next

Simplex of variance partitioning

```
mME_pool %>%
  left_join(mME %>%
              mutate(omics = 'ME-NMR') %>%
              filter(effect=='ran_pars') %>%
              filter(str_detect(group,'Resi')),
            by = c('met','category')) %>%
  ggplot(data = ., aes(`sd(logy)`,estimate)) +
  geom_point() + coord_equal() + geom_abline()
```

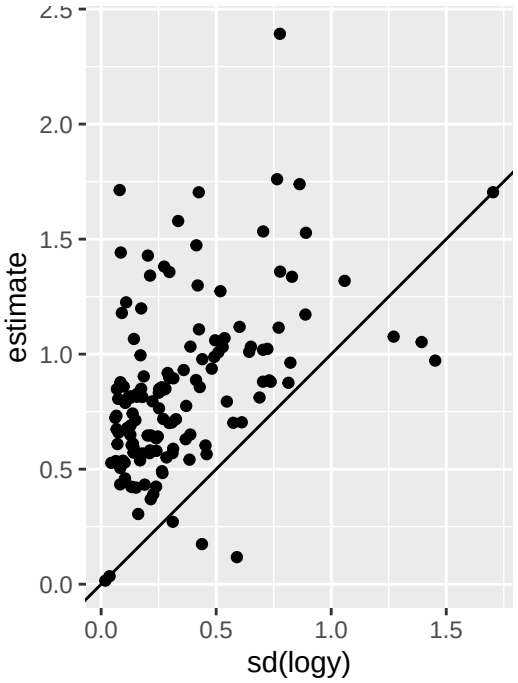


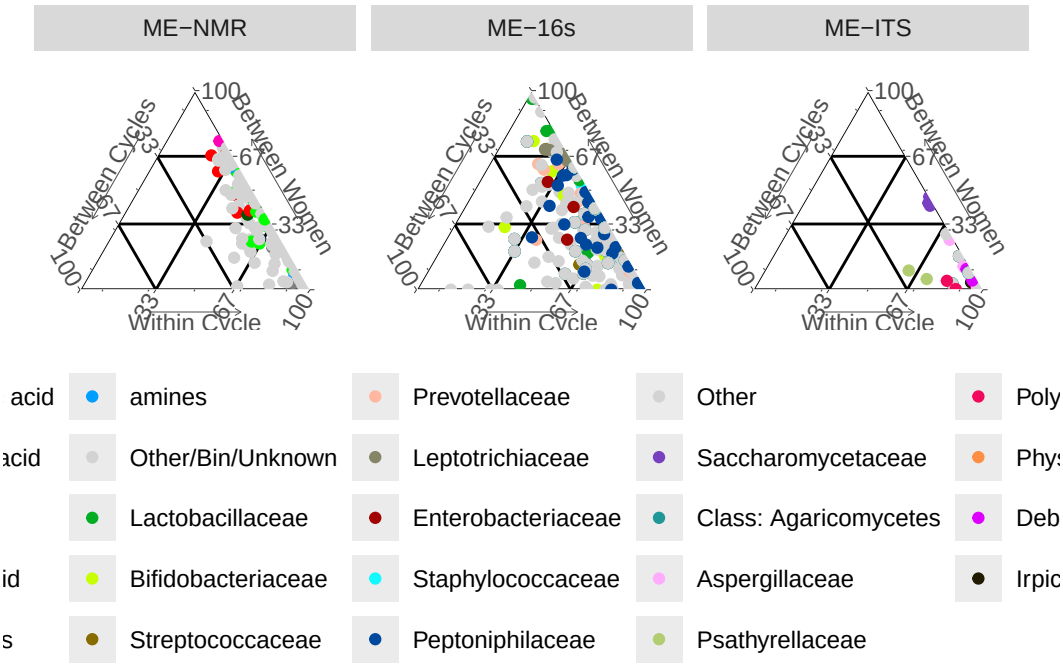
Figure 3A in publication

```
tbseq <- bind_rows(tbITS %>% mutate(omics = 'ME-ITS'),
                   tb16s %>% mutate(omics = 'ME-16s')) %>%
  filter(effect=='ran_pars') %>%
  dplyr::select(-p.value) %>%
  dplyr::mutate(vr = estimate^2) %>%
  group_by(omics,ta3,ta4,ta5,ta6,ta7,OTU) %>%
  dplyr::mutate(relvr = vr / sum(vr))
```

```
tbME <- mME %>%
  mutate(omics = 'ME-NMR') %>%
  filter(effect=='ran_pars') %>%
  dplyr::select(-p.value) %>%
  group_by(met,category) %>%
  dplyr::mutate(vr = estimate^2,
                relvr = vr / sum(vr))

brks <- c(0,0.33,0.67,1.00)
gSIMPLX_ME <- tbseq %>%
  dplyr::select(omics,group,relvr) %>%
  spread(group,relvr) %>%
  left_join(bind_rows(bsum16s,bsumITS), by = 'ta5') %>%
  bind_rows(
    tbME %>% filter(!str_detect(met,'PC')) %>%
      dplyr::select(omics,group,relvr) %>%
      spread(group,relvr) %>% left_join(dflb, by = 'category') ) %>%
  mutate(cycle_part = ifelse(cycle_part==0,0.02,cycle_part),
         Participant = Participant /(Participant + cycle_part + Residual),
         Residual = Residual /(Participant + cycle_part + Residual)) %>%
  ungroup %>%
  mutate(omics = factor(omics, levels = c('ME-NMR','ME-16s','ME-ITS'))) %>%
  ggtern(data =., aes(cycle_part, Participant, Residual, color = lb)) +
  geom_point() +
  scale_T_continuous(breaks = brks) +
  scale_L_continuous(breaks = brks) +
  scale_R_continuous(breaks = brks) +
  scale_color_manual(values = bind_rows(dflb,bsum16s,bsumITS)$color,
                     breaks = bind_rows(dflb,bsum16s,bsumITS)$lb ) +
  xlab('Between Cycles') +
  ylab('Between Women') +
  zlab('Within Cycle') +
  theme(tern.panel.background = element_rect(fill = "white"),
        tern.panel.grid.major =  element_line(color = "black"),
        tern.axis.arrow.show = TRUE,
        tern.axis.title.T = element_blank(),
        tern.axis.title.L = element_blank(),
        tern.axis.title.R = element_blank(),
        legend.position = 'bottom') +
  facet_wrap(~omics)

gSIMPLX_ME
```




```
ggsave('FIG3A_simplex.pdf',plot = gSIMPLX_ME, width = 7)
```